DSC 207 / PROJECT MILESTONE

Preparing LIGO data for inference across observing runs

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Motivation and current scope

We investigate whether a neural posterior estimator trained on O3a Hanford (H1) noise remains calibrated when applied to O4a noise. Fast parameter estimates are scientifically useful only when their uncertainty is reliable. Differences in detector noise and SNR-based selection can affect this reliability. This milestone covers dataset preparation, preprocessing validation and exploratory analysis toward the planned cross-run comparison.

Dataset and acquisition progress

The dataset consists of seven nonoverlapping 512-second H1 windows at 4,096 Hz from GWOSC: five O3a and two O4a windows. The accepted crops contain **3,584 seconds and 14,680,064 raw strain samples**. All seven passed metadata, finiteness, quality-flag and transient checks. The source files span 4,096 seconds each; the dataset sizes below count only the retained 512-second crops.

Raw split	Groups	Seconds	Strain samples
O3a training	3	1,536	6,291,456
O3a validation	1	512	2,097,152
O3a test	1	512	2,097,152
O4a test	2	1,024	4,194,304

Selection requires DATA and CBC_CAT1/CAT2 flags, no CBC or burst hardware injection, and a one-hour exclusion around cataloged events and injection gaps. EDA motivated an additional transient screen: reject a window if any eight-second block standard deviation exceeds three times the median block standard deviation. Of 36 O3a candidates, 26 failed interval selection and five failed this screen; five were retained. Both O4a candidates passed. The replacement validation crop starts at GPS **1249325120**. Rejected sources were replaced rather than imputed or clipped.

Simulation and present allocations

Labeled examples are generated with PyCBC IMRPhenomD waveforms, using component masses of 5-100 solar masses, aligned spins of 0-0.8 and Euclidean volume-weighted distances of 100-2,000 Mpc. Plus-polarization waveforms are peak-aligned and added to copied noise buffers. The four targets are chirp mass (solar masses), mass ratio, effective spin and distance (Mpc); mass ratio and spin are dimensionless. The plus-only convention simplifies the detector response.

The initial two-source simulation run produced **44 accepted O3a training injections and nine accepted O4a test injections** from 50 attempts per run. Validation and O3a-test injection generation will use the independently allocated source groups.

In total, **53 injections were accepted, 47 fell below SNR 8, and no generation failures were reported**. These examples use the first O3a and O4a sources, which were unchanged by the source cleanup. They are an initial subset, not the complete seven-source dataset. Final processed example counts and fitting/validation/test allocations will be measured after the full preparation chain.

Raw-data exploration and cleanup

Raw-data EDA includes full time series, one-second zooms, standardized amplitude distributions and eight-second block statistics. The following figures describe the accepted seven-source dataset. Distribution shape and within-window variation guide the quality screening and the choice of noise normalization.

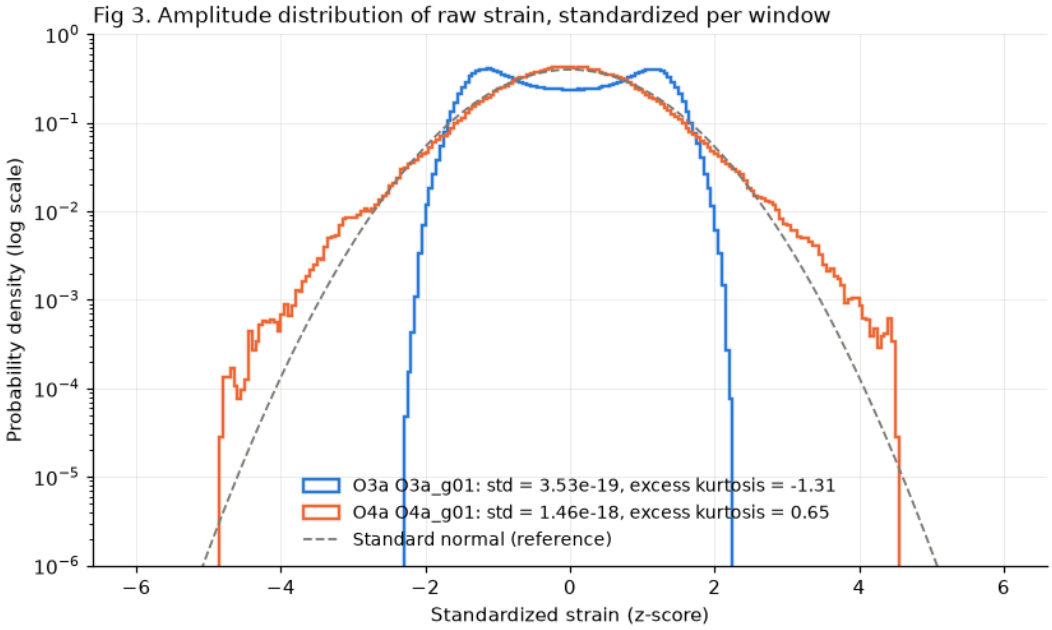


Figure 1. *O3a_g01 and O4a_g01 standardized strain with a Gaussian reference. Near-zero skew does not imply Gaussianity: excess kurtosis is about -1.312 and 0.646. Plot standardization is not model-input normalization.*

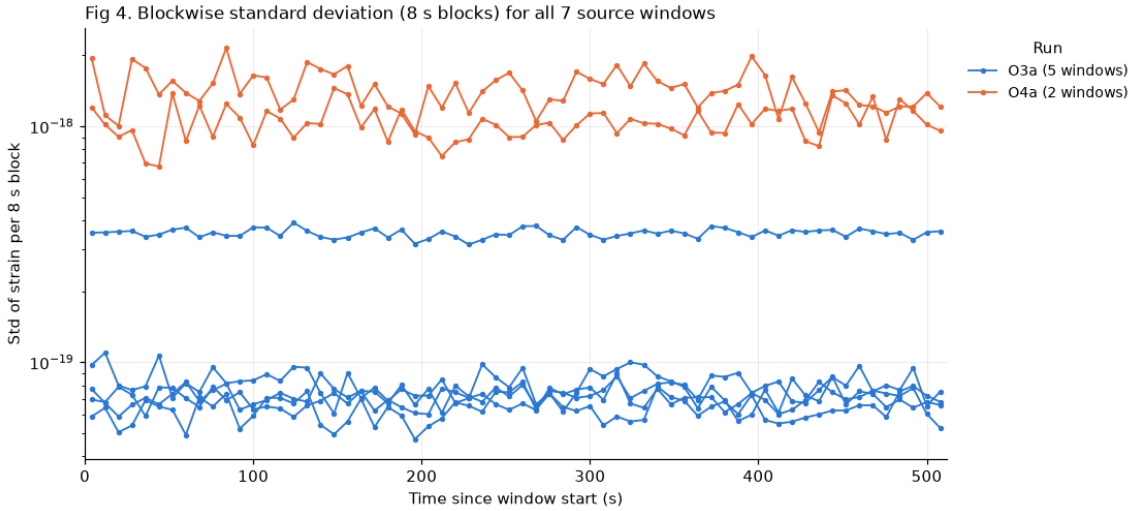


Figure 2. *Eight-second block std across seven windows. Max/min ratios range from 1.24-1.95 for O3a and 2.15-2.32 for O4a. Passing R5 does not make noise stationary.*

Interpretation and remaining input-target analysis

Median raw strain standard deviation is approximately 7.404×10^{-20} for O3a and 1.258×10^{-18} for O4a. This broadband statistic does not measure sensitivity within the analysis band. Some retained windows contain BURST_CAT2 vetoed seconds; acceptance follows the stated CBC-focused criteria rather than requiring every burst veto to pass.

The initial injection subset also has target-distribution, target-correlation and distance-SNR analyses. Waveform-target comparisons will use the processed inputs. Model-based importance will be assessed separately; correlation alone does not establish feature importance.

Preprocessing, next experiments and contributions

Preprocessing implementation and data challenges

The preprocessing chain estimates a background-only Welch PSD, applies a positive numerical floor, whitens in the frequency domain and filters with a fourth-order Butterworth 35-350 Hz bandpass using second-order sections and forward/backward application. Padded buffers are processed before extracting a four-second merger-centered window: 16,384 samples, with two seconds on each side. These processed samples form the model input. Controlled synthetic validation gave a maximum manual-versus-SciPy Welch relative error of 4.02×10^{-14} and reduced in-band PSD spread from 15.32 to 0.92 dB. Crops had exactly 16,384 finite samples, and under-padded buffers were rejected.

Numeric strain requires no categorical encoding; run, split and source-group identifiers remain metadata. No learned feature selection or per-example amplitude normalization is used, preserving amplitude information relevant to distance. Invalid or incorrectly shaped buffers are rejected, while valid loud signals are retained. Dataset checks cover observation-label IDs, source-group separation and overlapping crop windows. Current challenges are extending the initial injection subset, managing nonstationary noise and SNR selection, and completing processed-data accounting.

Methodology planned after dataset validation

Algorithm	Reason and next experiment
Linear regression - baseline	Four-target point estimates; compare with a training-mean reference.
Ridge - improvement 1	Regularize the same linear model. Select alpha from 0.1, 1, 10, 100, 1,000 on O3a validation.
SNPE + MAF - improvement 2	Learn a joint conditional posterior. Begin with a small run and recorded architecture/training settings.

Scalers will be fitted on training data only, with source groups separated across splits. Model settings will be selected on O3a validation and compared on identical evaluation examples using per-target MAE, RMSE and R-squared. Aggregate errors will use training-only target scales. SNPE trials will record architecture, batch size, learning rate, early stopping and seed. Evaluation will compare posterior means, central 50%, 80% and 90% coverage, interval widths and failure rates, using comparable posterior draw budgets. The preferred model will balance validation error with reliable coverage and useful interval widths.

Uniform component draws do not produce a uniform rectangular distribution over the derived targets, and SNR selection changes the population further. The simulator, inference prior and selection assumptions will therefore be aligned before formal SBC. The EDA-informed noise-screening rule will be frozen before model selection; fresh evaluation data will be reserved if subsequent tuning uses held-out outcomes. The present pilot does not establish calibration drift.

Contributions and notebook ownership

Member / notebook	Specific contributions
Kennedy Goliman / 01	Seven-source acquisition, EDA, transient-based source replacement and split/export checks.
Aaron Rassiq / 02	Parameter draws, waveforms, injection/SNR ledger and saved population EDA.
Noah Fonck / 03	PSD estimation, whitening, bandpass and crop implementation, with successful controlled synthetic validation.
Ali Lordifar / 04	Updated source/injection accounting, raw-EDA review, handoff checks, selection plots and methodology plan.

Notebook directory: notebooks/milestone_preprocessing_roles/. Each numbered notebook has one owner: 01_raw_data_provenance_and_eda.ipynb; 02_simulations_labels_and_selection.ipynb; 03_preprocessing_and_quality_checks.ipynb; 04_dataset_audit_eda_and_methodology_plan.ipynb.

Next stage

The next experiments will extend injection generation to all allocated source groups, apply the validated preprocessing chain and reconcile retained examples with their labels. Final processed split counts, waveform-target EDA and validation-based feature importance will be reported before the common model comparison.

Repository: github.com/alilordifar/ligo-parameter-inference
Data source: GWOSC public strain releases. Preprocessing validation results refer to controlled synthetic inputs; raw-data EDA and simulation counts describe the manual pilot.